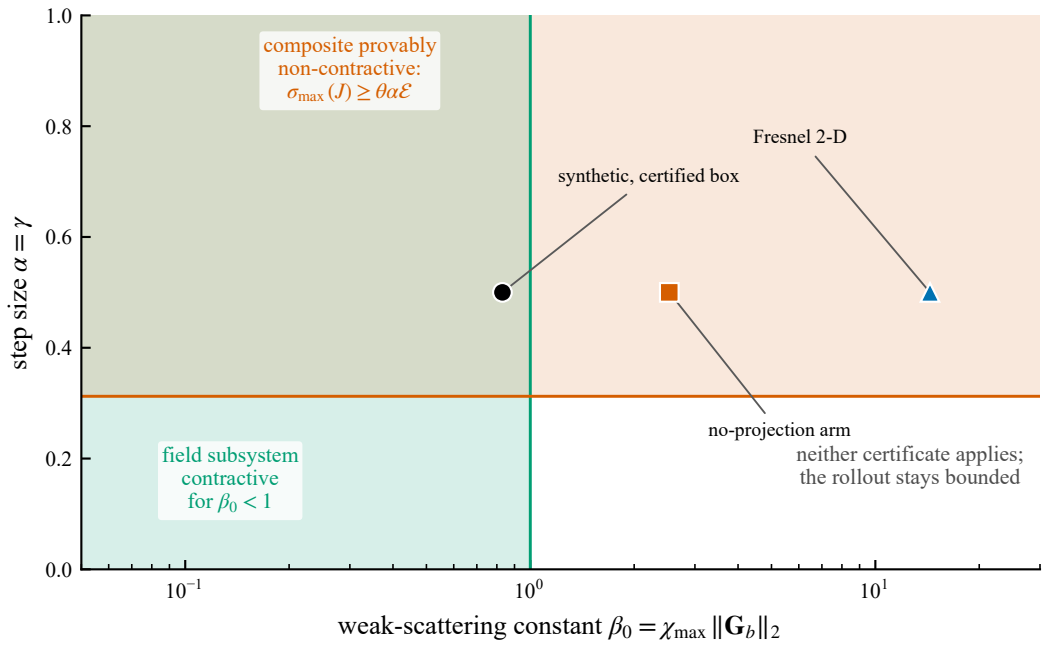
composite provably
non-contractive:
$\sigma_{\max}(J) \geq \theta\alpha\mathcal{E}$
Fresnel 2-D
synthetic, certified box
no-projection arm
neither certificate applies;
the rollout stays bounded
field subsystem
contractive
for $\beta_0 < 1$
step size $\alpha = \gamma$
weak-scattering constant $\beta_0 = \chi_{\max}\|\mathbf{G}_b\|_2$
1.0
0.8
0.6
0.4
0.2
0.0
$10^{-1}$
$10^{0}$
$10^{1}$